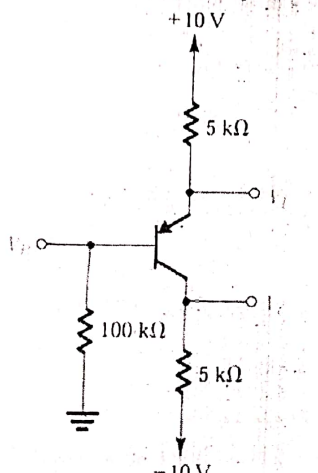


Degree: B.Tech ECAM Semester: 3rd Sem
END-SEMESTER EXAMINATION, JULY 2026
 Course Title: Microelectronics Circuits and Applications
 Course Code: EAEPC304

Duration: 3:00 Hours

Max. Marks: 50

Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

S. No.	Question	Marks	CO
Q1	Attempt any two parts of the following		
1a	Using the small-signal model of a MOSFET, derive the expression for the voltage gain of a Common-Source (CS) amplifier.	5	CO1
1b	Explain source degeneration in a Common-Source amplifier. Derive the gain expression and discuss its advantages.	5	CO1
1c	In the circuit shown below, measurement indicates V_B to be +1.0 V and V_E to be +1.7 V. What are α and β for this transistor? What voltage V_C do you expect at the collector?	5	CO1
			
Q2	Attempt any two parts of the following		
2a	Explain the DC operation of a MOS Differential Pair. Derive the drain current expressions when both inputs are equal.	5	CO2
2b	Explain the effect of transistor mismatch on the performance of a differential amplifier.	5	CO2
2c	A MOS differential pair is operated at a total bias current of 0.8mA, using transistors with W/L ratio of 100, $\mu_n C_{ox} = 0.2 \text{ mA/V}^2$, $V_A = 20\text{V}$ and $R_D = 5\text{k}\Omega$. Find V_{ov} , g_m , r_o and A_d .	5	CO2
Q3	Attempt any two parts of the following		
3a	A square-wave signal is applied to an Op-Amp circuit and a triangular waveform is obtained at the output. Identify the circuit used and explain the reason for the waveform transformation.	5	CO3
3b	A signal processing system requires the output to represent the rate of change of the input signal. Which Op-Amp circuit would you recommend? Explain its operation and discuss practical limitations.	5	CO3

3c	Design a noninverting amplifier with a gain of 2. At the maximum output voltage of 10 V, the current in the voltage divider is to be 10 μ A.	5	CO3
Q4	Attempt any two parts of the following		
4a	Why is a 555 timer often referred to as a "universal timing IC"? Justify the statement by discussing at least three different modes/applications of operation.	5	CO4
4b	Explain the operation of a logarithmic amplifier and derive the expression for its output voltage.	5	CO4
4c	How can a square-wave generator be implemented using an Op-Amp? Explain with a circuit diagram.	5	CO4
Q5	Attempt any two parts of the following		
5a	Explain the properties of negative feedback in amplifiers. How does negative feedback affect gain, bandwidth, noise and distortion?	5	CO5
5b	A designer wants to increase the bandwidth and improve the stability of an amplifier without changing the active device. What technique would you recommend? Explain its advantages and limitations.	5	CO5
5c	Compare Class A, Class B, and Class AB power amplifiers in terms of efficiency, conduction angle, and distortion.	5	CO5